

–Appendix–

Prune but Preserve: Decoupling Redundancy Suppression and Semantic Preservation for Efficient Time Series Clustering

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I. BASELINE DESCRIPTIONS

The compared methods are selected according to the main empirical question of the paper. Since SIGMA studies whether redundancy should be controlled before encoder-side feature interaction, the baselines need to cover full-token participation, input-token exclusion, partial-observation modeling, structural matching, and general time series representation learning. This organization makes the comparison aligned with the prune-and-preserve dilemma discussed in the main paper. For methods that output embeddings, k -means with the label-provided cluster number is applied for final cluster assignment, following the protocol in the main paper.

FCACC [1] is a fuzzy cluster-aware contrastive clustering method for time series. The method constructs augmented views and introduces cluster-aware hard negatives to reduce the mismatch between instance-level contrast and cluster-level assignment. Since FCACC refines cluster structure mainly after sequence representation has been formed, it represents a clustering-oriented full-participation baseline without source-level patch admission. Therefore, FCACC tests whether cluster-aware contrast alone can overcome background-dominated feature interaction. The official implementation is available at [FCACC](#).

TFMCC [2] is a time-frequency augmented multi-level contrastive clustering method for time series. The method builds augmented views from temporal and spectral perspectives and applies contrastive objectives at multiple representation levels. TFMCC keeps the complete input sequence in the representation pipeline and strengthens useful patterns through time-frequency augmentation and contrastive refinement. Therefore, TFMCC provides a representative full-token participation baseline for evaluating whether SIGMA brings additional gains by reducing redundant patch participation before feature interaction. The official implementation is available at [TFMCC](#).

GTM [3] is a general time series representation model for heterogeneous downstream tasks. The model learns temporal-granularity-aware representations through frequency-domain

attention and shared sequence modeling. Since GTM is designed as a general representation learner instead of a clustering-specific redundancy controller, k -means is applied to its learned embeddings for final assignment. GTM therefore helps examine whether broad temporal pretraining can replace explicit source-level admission in redundancy-dominated MTS clustering. The official implementation is available at [GTM](#).

EMTC [4] is an evolving-masked representation learning framework for MTS clustering. The method suppresses redundancy through Importance-aware Variate-wise Masking and organizes retained content by Multi-Endogenous Views. Its dual-path learning combines Consistency and Reconstruction Learning with Clustering-guided MEV Contrastive learning. EMTC is therefore a strong input-token exclusion baseline, since it already reduces token participation before clustering-oriented representation learning. Comparing SIGMA with EMTC further tests whether confidence-calibrated admission, routed semantic restoration, and prototype-guided refinement provide stronger prune-and-preserve behavior after token exclusion. The official implementation is available at [EMTC](#).

FEI [5] is a non-contrastive representation learning method based on frequency-masked embedding inference. The method masks frequency components and predicts the corresponding embeddings, encouraging the representation to capture informative spectral structures without manually designed positive and negative pairs. FEI is relevant to the redundancy discussion because it controls information in the frequency domain, but it does not decide which input patches can enter encoder-side interaction. Thus, FEI provides a complementary spectral-masking baseline for assessing whether source-level patch admission is still beneficial beyond frequency-oriented representation learning. The official implementation is available at [FEI](#).

MVCIMTS [6] is a contrastive multi-view clustering method for incomplete MTS. The method treats variates as different views and learns complementary information through multi-view contrastive learning. MVCIMTS is related to partial-observation learning, but its incompleteness comes from variable-level views instead of patch-level source admis-

sion. Therefore, MVCIMTS tests whether multi-view contrast over incomplete observations can provide the same clustering benefit as SIGMA, which prunes redundant patches first and then restores semantic content through routed objectives. The official implementation is available at [MVCIMTS](#).

***k*-Graph** [7] is a graph-based time series clustering method. The method converts subsequences into graph representations and performs clustering on graph-derived features. By constructing graphs over multiple subsequence lengths, *k*-Graph captures recurring local patterns and provides interpretable structural descriptions for clusters. It differs from full-token neural encoders and input-token exclusion methods because representation construction relies on explicit graph summaries. Hence, *k*-Graph serves as a structural baseline for datasets where shape-level or recurrence-level similarity dominates the clustering criterion. The official implementation is available at [k-Graph](#).

TimesURL [8] is a self-supervised contrastive method for universal time series representation learning. The method learns general temporal embeddings by combining contrastive objectives with constraints tailored to time series data. TimesURL is included to examine whether universal self-supervised representations can provide stable clustering embeddings without an explicit admission mechanism. In the context of the prune-and-preserve dilemma, TimesURL represents a general full-sequence representation baseline whose redundancy control is not performed at the input-patch level. The official implementation is available at [TimesURL](#).

UNITS [9] is a unified multi-task time series model. It represents task information as tokens within a shared modeling framework and uses a modified Transformer architecture to learn transferable representations across heterogeneous domains, temporal scales, and sampling patterns. UNITS provides a foundation-style baseline for assessing whether task-unified pretraining is sufficient for redundancy-dominated MTS clustering. Since UNITS does not explicitly restrict background-dominated patch participation before encoding, its comparison with SIGMA highlights the role of source-level admission and routed preservation in cluster-oriented representation learning. The official implementation is available at [UNITS](#).

A. Implementation Details

All experiments follow the unsupervised clustering protocol in the main paper. For each UEA dataset, the training and test splits are concatenated before clustering, and labels are accessed only for evaluation. The argument `dataname` specifies the dataset to run. Its default value only provides an executable entry for the script and does not indicate a special benchmark setting. Each experiment is repeated with five random seeds, namely 2022, 2023, 2024, 2025, and 2026. The reported results are the mean and standard deviation over the five runs.

The default SIGMA setting enables Confidence-Calibrated Masked Admission by setting `use_mask=1`. For the corresponding masking ablation, `use_mask=0` removes source-

level admission and keeps the remaining configuration unchanged. This comparison isolates the effect of admission control before encoder-side feature interaction. Unless otherwise specified, the input noise level is set to 0. Noise perturbation is introduced only in the robustness study.

For patch-level admission, the patch length is set to $\ell_p = 4$. Given $m = \lceil t/\ell_p \rceil$ patches, the retained ratio is set to $r_a = 0.5$, and the retained patch number follows $m_a = \lceil r_a m \rceil$. Therefore, the default encoder input contains half of the patch sequence after admission. This setting follows the prune-and-preserve principle, where redundant patch participation is reduced before feature interaction and the retained content remains sufficient for semantic restoration.

For multi-view representation learning, the number of endogenous views is set to $n_v = 2$. Each view encoder uses hidden dimension $d_h = 32$, and the fused clustering embedding uses representation dimension $d_z = 64$. The prototype alignment temperature is set to $\tau_p = 0.1$. These values keep the representation compact and provide enough capacity for endogenous view organization. In robustness experiments, Gaussian perturbation is controlled by the noise standard deviation, and the default setting uses no added perturbation.

SIGMA is trained for $e_{\max} = 200$ epochs with mini-batch size $b = 64$. Adam is adopted with learning rate $\alpha = 0.001$. The loss weights are set as $\lambda_{\text{rec}} = 1.0$, $\lambda_{\text{orth}} = 0.1$, and $\lambda_{\text{pa}} = 1.0$. Here, λ_{rec} controls the reconstruction objective $\mathcal{L}_{\text{rec}}$, λ_{orth} controls the inter-view orthogonality objective $\mathcal{L}_{\text{orth}}$, and λ_{pa} controls the prototype alignment objective $\mathcal{L}_{\text{pa}}$. The reconstruction term anchors retained temporal content, the orthogonality term encourages complementary endogenous views, and prototype alignment refines the fused cluster space. Under this configuration, admission, semantic preservation, and cluster refinement remain assigned to distinct roles within the decoupled optimization framework.

II. SUPPLEMENTARY RESULTS

This section extends the empirical analysis in the main paper. The main paper reports representative per-dataset results and condensed diagnostic studies to keep the experimental narrative focused. To make the empirical claims fully traceable, the appendix provides complete benchmark results, full ablation results, additional training-loss curves and industrial-scale efficiency tests. All supplementary studies follow the protocol defined in Sec. I.

The results are organized around the central question of SIGMA, namely whether source-level admission can reduce redundant feature interaction and whether routed objectives can preserve semantic content after partial observation. Complete clustering tables examine the overall benchmark behavior. Complete ablations verify the role of each module and route. Training-loss curves inspect the staged optimization process. Efficiency tests show how increasing mask rates reduce encoder-side cost under large-scale settings. Together, these analyses connect redundancy suppression, semantic preservation, and cluster refinement within the SIGMA framework.

TABLE I
COMPLETE CLUSTERING RESULTS ON 30 UEA DATASETS EVALUATED BY ACC, F1, NMI, AND ARI, PART I. THE COMPLETE RESULTS ON ALL DATASETS ARE PROVIDED IN THE ONLINE APPENDIX. THE **BEST** AND **SECOND BEST** RESULTS ARE HIGHLIGHTED BY CELL BACKGROUND.

Datasets	Metric	FCACC (PR'26)	GTM (ICLR'26)	EMTC (AAAI'26)	TFMCC (AAAI'26)	FEI (AAAI'25)	MVCIMTS (INFFUS'25)	k-Graph (TKDE'25)	TimesURL (AAAI'24)	UNITS (NeurIPS'24)	SIGMA (ours)
ArticulatoryWordRecognition	ACC	0.7329 ± 0.0037	0.3555 ± 0.0380	0.3547 ± 0.0185	0.5030 ± 0.0253	0.4710 ± 0.0731	0.0497 ± 0.0124	0.2838 ± 0.0159	0.6929 ± 0.0516	0.2948 ± 0.0200	0.7603 ± 0.0232
	F1	0.8039 ± 0.0055	0.3526 ± 0.0348	0.3560 ± 0.0214	0.3994 ± 0.0241	0.4974 ± 0.0815	0.0109 ± 0.0096	0.2774 ± 0.0169	0.7425 ± 0.0413	0.3114 ± 0.0243	0.7929 ± 0.0392
	NMI	0.8408 ± 0.0064	0.4708 ± 0.0311	0.5070 ± 0.0272	0.7012 ± 0.0159	0.6185 ± 0.0744	0.0314 ± 0.0428	0.3948 ± 0.0099	0.8215 ± 0.0145	0.4280 ± 0.0198	0.8276 ± 0.0165
	ARI	0.6641 ± 0.0045	0.1833 ± 0.0296	0.1615 ± 0.0271	0.4466 ± 0.0210	0.3352 ± 0.0843	0.0019 ± 0.0031	0.1224 ± 0.0091	0.6214 ± 0.0340	0.1395 ± 0.0188	0.6466 ± 0.0257
AtrialFibrillation	ACC	0.4333 ± 0.0000	0.4667 ± 0.0422	0.6133 ± 0.0499	0.4333 ± 0.0558	0.3867 ± 0.0163	0.4095 ± 0.0440	0.4400 ± 0.0389	0.4600 ± 0.0490	0.4500 ± 0.0167	0.5000 ± 0.0365
	F1	0.4303 ± 0.0028	0.4593 ± 0.0393	0.5828 ± 0.0633	0.4041 ± 0.0191	0.3543 ± 0.0194	0.3491 ± 0.0555	0.4295 ± 0.0406	0.4169 ± 0.0278	0.4343 ± 0.0089	0.4956 ± 0.0778
	NMI	0.1005 ± 0.0040	0.0726 ± 0.0277	0.2883 ± 0.0758	0.0544 ± 0.0351	0.0353 ± 0.0237	0.1445 ± 0.0821	0.1143 ± 0.0748	0.0858 ± 0.0339	0.0643 ± 0.0072	0.1251 ± 0.0390
	ARI	-0.0056 ± 0.0011	0.0051 ± 0.0258	0.1248 ± 0.0788	0.0064 ± 0.0282	-0.0360 ± 0.0089	0.1058 ± 0.0600	0.0039 ± 0.0462	0.0136 ± 0.0342	-0.0125 ± 0.0012	0.0582 ± 0.0393
BasicMotions	ACC	0.6675 ± 0.0100	0.7300 ± 0.1086	0.7950 ± 0.0292	0.9850 ± 0.0122	0.6775 ± 0.0443	0.2975 ± 0.0827	0.8875 ± 0.13121	0.7700 ± 0.1582	0.5437 ± 0.0812	0.9875 ± 0.0079
	F1	0.8228 ± 0.0098	0.7311 ± 0.1060	0.7832 ± 0.0275	0.9701 ± 0.0235	0.7351 ± 0.0723	0.1517 ± 0.0794	0.8792 ± 0.1500	0.8627 ± 0.0835	0.5705 ± 0.0412	0.9875 ± 0.0079
	NMI	0.8048 ± 0.0020	0.5569 ± 0.1633	0.6949 ± 0.0711	0.9627 ± 0.0248	0.6792 ± 0.0317	0.1007 ± 0.1775	0.8626 ± 0.0575	0.8487 ± 0.0634	0.3898 ± 0.1107	0.9865 ± 0.0191
	ARI	0.6295 ± 0.0045	0.4526 ± 0.1902	0.6085 ± 0.0606	0.9606 ± 0.0311	0.5317 ± 0.0351	0.0690 ± 0.1381	0.8174 ± 0.1218	0.7349 ± 0.1448	0.2989 ± 0.1276	0.9666 ± 0.0208
CharacterTrajectories	ACC	0.7981 ± 0.0012	0.0051 ± 0.0000	0.3000 ± 0.0261	0.3677 ± 0.0448	0.6040 ± 0.0096	0.0866 ± 0.0318	0.4500 ± 0.0167	0.0651 ± 0.0000	0.4533 ± 0.0267	0.7668 ± 0.0529
	F1	0.8583 ± 0.0012	0.0061 ± 0.0000	0.2886 ± 0.0434	0.3405 ± 0.0281	0.6289 ± 0.0099	0.0196 ± 0.0166	0.4350 ± 0.0149	0.1222 ± 0.0000	0.1703 ± 0.0373	0.8019 ± 0.0378
	NMI	0.8726 ± 0.0013	0.0000 ± 0.0000	0.3633 ± 0.0432	0.5756 ± 0.0418	0.7095 ± 0.0041	0.0682 ± 0.1107	0.5347 ± 0.0121	0.0000 ± 0.0000	0.1776 ± 0.0688	0.8261 ± 0.0315
	ARI	0.7705 ± 0.0012	0.0000 ± 0.0000	0.1756 ± 0.0296	0.1913 ± 0.0474	0.5137 ± 0.0138	0.0181 ± 0.0345	0.3211 ± 0.0193	0.0000 ± 0.0000	0.0532 ± 0.0279	0.6933 ± 0.0563
Cricket	ACC	0.9407 ± 0.0288	0.4533 ± 0.0264	0.5528 ± 0.0204	0.6922 ± 0.0664	0.5744 ± 0.0834	0.1167 ± 0.0350	0.4589 ± 0.0218	0.6722 ± 0.0813	0.4922 ± 0.0990	0.8489 ± 0.0787
	F1	0.9599 ± 0.0024	0.4393 ± 0.0376	0.5635 ± 0.0383	0.6319 ± 0.0703	0.6027 ± 0.0590	0.0417 ± 0.0250	0.4545 ± 0.0225	0.7562 ± 0.0557	0.5414 ± 0.0992	0.8838 ± 0.0523
	NMI	0.9509 ± 0.0020	0.5189 ± 0.0101	0.6181 ± 0.0286	0.7858 ± 0.0464	0.6794 ± 0.0579	0.0817 ± 0.0964	0.4752 ± 0.0228	0.7958 ± 0.0391	0.5947 ± 0.0841	0.8764 ± 0.0455
	ARI	0.9051 ± 0.0158	0.2584 ± 0.0194	0.2971 ± 0.0365	0.5842 ± 0.0827	0.4468 ± 0.0872	0.0171 ± 0.0299	0.2533 ± 0.0250	0.6028 ± 0.0783	0.3359 ± 0.1072	0.7672 ± 0.0967
DuckDuckGeese	ACC	0.3067 ± 0.0125	0.4140 ± 0.0162	0.4240 ± 0.0408	0.4720 ± 0.0172	0.3220 ± 0.0117	0.2620 ± 0.0519	0.3280 ± 0.0160	0.3220 ± 0.0147	0.2960 ± 0.0136	0.4020 ± 0.0325
	F1	0.2611 ± 0.0200	0.4042 ± 0.0246	0.4003 ± 0.0494	0.3394 ± 0.0193	0.3013 ± 0.0347	0.1650 ± 0.0575	0.3017 ± 0.0107	0.2641 ± 0.0182	0.2869 ± 0.0075	0.3819 ± 0.0291
	NMI	0.0800 ± 0.0179	0.2483 ± 0.0205	0.2082 ± 0.0380	0.2791 ± 0.0267	0.1030 ± 0.0990	0.0335 ± 0.0301	0.0958 ± 0.0101	0.1732 ± 0.0351	0.0682 ± 0.0175	0.1718 ± 0.0577
	ARI	0.0227 ± 0.0109	0.1394 ± 0.0210	0.0729 ± 0.0331	0.1780 ± 0.0248	0.0369 ± 0.0091	0.0073 ± 0.0170	0.0225 ± 0.0087	0.0375 ± 0.0110	0.0064 ± 0.0105	0.1000 ± 0.0580
EigenWorms	ACC	OOM	0.3876 ± 0.0072	OOM	OOM	0.4185 ± 0.0361	0.4324 ± 0.0024	0.3135 ± 0.0236	0.3660 ± 0.0510	0.4525 ± 0.0366	0.4332 ± 0.0402
	F1	OOM	0.2868 ± 0.0068	OOM	OOM	0.3899 ± 0.0605	0.1545 ± 0.0173	0.2920 ± 0.0180	0.3035 ± 0.0806	0.4410 ± 0.0601	0.4016 ± 0.0483
	NMI	OOM	0.0083 ± 0.0035	OOM	OOM	0.1826 ± 0.0152	0.0300 ± 0.0094	0.0848 ± 0.0181	0.0697 ± 0.0572	0.2872 ± 0.0919	0.8185 ± 0.0300
	ARI	OOM	0.0626 ± 0.0052	OOM	OOM	0.1453 ± 0.0227	0.0098 ± 0.0046	0.0400 ± 0.0160	0.0515 ± 0.0699	0.1694 ± 0.0486	0.1620 ± 0.0458
Epilepsy	ACC	0.4279 ± 0.0297	0.3869 ± 0.0404	0.5203 ± 0.0244	0.4953 ± 0.0477	0.6364 ± 0.0808	0.2953 ± 0.0324	0.6167 ± 0.0316	0.5156 ± 0.0189	0.4982 ± 0.0646	0.4756 ± 0.0321
	F1	0.4295 ± 0.0302	0.3710 ± 0.0430	0.5136 ± 0.0265	0.4147 ± 0.0552	0.6200 ± 0.0821	0.1882 ± 0.0235	0.6200 ± 0.0260	0.4979 ± 0.0080	0.5086 ± 0.0930	0.4724 ± 0.0357
	NMI	0.1397 ± 0.0347	0.1015 ± 0.0345	0.2476 ± 0.0419	0.3029 ± 0.0794	0.4079 ± 0.0773	0.0205 ± 0.0340	0.4013 ± 0.0415	0.3363 ± 0.0795	0.3126 ± 0.0851	0.2652 ± 0.0289
	ARI	0.0960 ± 0.0211	0.0814 ± 0.0434	0.1803 ± 0.0389	0.1992 ± 0.0638	0.3829 ± 0.0804	0.0076 ± 0.0184	0.3275 ± 0.0526	0.2288 ± 0.0542	0.2320 ± 0.0733	0.1889 ± 0.0197
ERing	ACC	0.6327 ± 0.0360	0.4180 ± 0.0585	0.6704 ± 0.0486	0.8607 ± 0.0218	0.7120 ± 0.0918	0.3667 ± 0.1560	0.5380 ± 0.0421	0.7433 ± 0.1161	0.4087 ± 0.0639	0.7980 ± 0.0749
	F1	0.6306 ± 0.0393	0.4117 ± 0.0586	0.6741 ± 0.0496	0.7564 ± 0.0332	0.7508 ± 0.0811	0.2682 ± 0.1646	0.5469 ± 0.0403	0.7862 ± 0.0839	0.4159 ± 0.0698	0.8003 ± 0.0701
	NMI	0.5660 ± 0.0104	0.2357 ± 0.0481	0.5487 ± 0.0473	0.7821 ± 0.0321	0.6562 ± 0.0772	0.3054 ± 0.2115	0.3843 ± 0.0143	0.7407 ± 0.0313	0.2528 ± 0.0802	0.7672 ± 0.0892
	ARI	0.4231 ± 0.0204	0.1602 ± 0.0466	0.4382 ± 0.0499	0.7051 ± 0.0415	0.5609 ± 0.0959	0.1897 ± 0.1599	0.2816 ± 0.0319	0.6467 ± 0.0757	0.1578 ± 0.0647	0.6168 ± 0.1270
EthanolConcentration	ACC	0.2992 ± 0.0025	0.2786 ± 0.0051	0.3605 ± 0.0224	0.2634 ± 0.0096	0.2901 ± 0.0017	0.2962 ± 0.0172	0.2920 ± 0.0100	0.2662 ± 0.0044	0.2691 ± 0.0034	0.2969 ± 0.0153
	F1	0.2759 ± 0.0027	0.2529 ± 0.0093	0.3458 ± 0.0279	0.4681 ± 0.0189	0.2744 ± 0.0087	0.2442 ± 0.0306	0.2882 ± 0.0110	0.2747 ± 0.0067	0.2090 ± 0.0059	0.2732 ± 0.0102
	NMI	0.0149 ± 0.0009	0.0047 ± 0.0010	0.0574 ± 0.0189	0.0124 ± 0.0098	0.0116 ± 0.0018	0.0176 ± 0.0112	0.0074 ± 0.0029	0.0199 ± 0.0025	0.0025 ± 0.0015	0.0190 ± 0.0157
	ARI	0.0012 ± 0.0003	-0.0013 ± 0.0010	0.0307 ± 0.0121	0.0003 ± 0.0007	0.0032 ± 0.0005	0.0072 ± 0.0073	0.0008 ± 0.0024	0.0039 ± 0.0009	-0.0014 ± 0.0010	0.0097 ± 0.0119
FaceDetection	ACC	0.5130 ± 0.0030	OOM	0.5146 ± 0.0047	0.5051 ± 0.0029	0.5077 ± 0.0031	OOM	0.5038 ± 0.0018	0.5001 ± 0.0000	0.5054 ± 0.0040	0.5198 ± 0.0044
	F1	0.5125 ± 0.0028	OOM	0.4802 ± 0.0369	0.4813 ± 0.0585	0.5051 ± 0.0039	OOM	0.5028 ± 0.0023	0.3336 ± 0.0000	0.5030 ± 0.0060	0.5192 ± 0.0046
	NMI	0.0005 ± 0.0002	OOM	0.0010 ± 0.0004	0.0002 ± 0.0002	0.0002 ± 0.0001	OOM	0.0001 ± 0.0000	0.0027 ± 0.0000	0.0001 ± 0.0002	0.0012 ± 0.0005
	ARI	0.0006 ± 0.0003	OOM	0.0007 ± 0.0006	0.0001 ± 0.0001	0.0002 ± 0.0002	OOM	-0.0000 ± 0.0000	0.0000 ± 0.0000	0.0001 ± 0.0003	0.0015 ± 0.0006
FingerMovements	ACC	0.5250 ± 0.0052	0.5183 ± 0.0085	0.5680 ± 0.0160	0.5072 ± 0.0055	0.5163 ± 0.0127	0.5260 ± 0.0041	0.5264 ± 0.0026	0.5192 ± 0.0082	0.5130 ± 0.0054	0.5221 ± 0.0129
	F1	0.4951 ± 0.0111	0.4982 ± 0.0212	0.5349 ± 0.0218	0.6004 ± 0.0599	0.5051 ± 0.0184	0.5230 ± 0.0023	0.4997 ± 0.0023	0.5161 ± 0.0058	0.5124 ± 0.0051	0.5183 ± 0.0074
	NMI	0.0027 ± 0.0009	0.0014 ± 0.0009	0.0210 ± 0.0132	0.0023 ± 0.0035	0.0013 ± 0.0016	0.0021 ± 0.0007	0.0028 ± 0.0005	0.0013 ± 0.0009	0.0006 ± 0.0003	0.0022 ± 0.0026
	ARI	0.0008 ± 0.0009	-0.0004 ± 0.0009	0.0121 ± 0.0087	-0.0010 ± 0.0008	-0.0005 ± 0.0021	0.0004 ± 0.0009	0.0009 ± 0.0005	-0.0006 ± 0.0013	-0.0016 ± 0.0004	0.0003 ± 0.0031
HandMovementDirection	ACC	0.2829 ± 0.0063	0.2957 ± 0.0087	0.4405 ± 0.0359	0.3248 ± 0.0164	0.3017 ± 0.0088	0.3017 ± 0.0123	0.3026 ± 0.0099	0.3120 ± 0.0097	0.3743 ± 0.0375	0.3179 ± 0.0100
	F1	0.2108 ± 0.0397	0.2907 ± 0.0061	0.4070 ± 0.0226	0.3328 ± 0.0189	0.2867 ± 0.0112	0.2667 ± 0.0205	0.2996 ± 0.0128	0.3089 ± 0.0099	0.3667 ± 0.0383	0.3145 ± 0.0104
	NMI	0.0168 ± 0.0104	0.0156 ± 0.0061	0.0910 ± 0.0245	0.0286 ± 0.0110	0.0184 ± 0.0049	0.0206 ± 0.0103	0.0144 ± 0.0047	0.0208 ± 0.0045	0.0818 ± 0.0394	0.0208 ± 0.0050
	ARI	-0.0047 ± 0.0038	0.0015 ± 0.0062	0.0475 ± 0.0166	0.0109 ± 0.0104	0.0001 ± 0.0045	0.0007 ± 0.0032	0.0003 ± 0.0032	0.0046 ± 0.0035	0.0488 ± 0.0282	0.0052 ± 0.0047
Handwriting	ACC	0.2433 ± 0.0041	0.1264 ± 0.0027	0.1536 ± 0.0097	0.1090 ± 0.0209	0.1778 ± 0.0097	0.0878 ± 0.0160	0.1708 ± 0.0101	0.1488 ± 0.0067	0.2940 ± 0.0068	0.2530 ± 0.0126
	F1	0.2418 ± 0.0080	0.1260 ± 0.0020	0.1501 ± 0.0114	0.1206 ± 0.0171	0.1741 ± 0.0071	0.0352 ± 0.0127	0.1679 ± 0.0098	0.1484 ± 0.0078	0.2881 ± 0.0075	0.2591 ± 0.0132
	NMI	0.3723 ± 0.0061	0.1523 ± 0.0052	0.1934 ± 0.0070	0.1228 ± 0.0510	0.2497 ± 0.0235	0.0637 ± 0.0274	0.2260 ± 0.0097	0.2009 ± 0.0035	0.0101 ± 0.0049	0.3652 ± 0.0139
	ARI	0.1106 ± 0.0035	0.0125 ± 0.0016	0.0254 ± 0.0036	0.0195 ± 0.0090	0.0494 ± 0.0107	0.0084 ± 0.0048	0.0429 ± 0.0045	0.0282 ± 0.0012	-0.0047 ± 0.0030	0.1134 ± 0.0113

A. Complete Clustering Results on 30 UEA Datasets

The main paper answers Q1 by combining representative per-dataset results with the Bonferroni-Dunn rank test. The purpose of this appendix subsection is to make that answer fully traceable at the dataset level. TABLE I and TABLE II report the complete clustering results on all 30 UEA datasets under ACC, F1, NMI, and ARI. These two tables contain the representative datasets discussed in the main paper and the remaining datasets omitted there due to space limits. Therefore, the appendix tables provide the full benchmark record behind Q1, not an additional subset comparison.

The complete results first confirm that the overall advantage of SIGMA does not come from a few favorable datasets. On datasets with clear cluster-revealing patterns, including ArticulatoryWordRecognition, BasicMotions, JapaneseVowels, PenDigits, and SelfRegulationSCP1, SIGMA obtains leading or highly competitive scores across multiple metrics. These datasets are important for evaluating the prune-and-preserve claim because sparse class-related variations can be diluted

when full sequences enter feature interaction. SIGMA reduces such background participation through source-level admission and then organizes the retained content through routed optimization. As a result, compact retained representations remain sufficient for clustering when discriminative temporal changes are sparse but informative. The results on JapaneseVowels and PenDigits are typical cases,

TABLE II
COMPLETE CLUSTERING RESULTS ON 30 UEA DATASETS EVALUATED BY ACC, F1, NMI, AND ARI, PART II. THE **BEST** AND **SECOND BEST** RESULTS ARE HIGHLIGHTED BY CELL BACKGROUND.

Datasets	Metric	FCACC (PR'26)	GTM (ICLR'26)	EMTC (AAAI'26)	TFMCC (AAAI'26)	FEI (AAAI'25)	MVCIMTS (INFLUS'25)	k-Graph (TKDE'25)	TimesURL (AAAI'24)	UNITS (NeurIPS'24)	SIGMA (ours)
Heartbeat	ACC	0.5844 ± 0.0079	0.5340 ± 0.0292	0.7180 ± 0.0020	0.7076 ± 0.0476	0.7090 ± 0.0000	0.6758 ± 0.0497	0.7296 ± 0.0025	0.7149 ± 0.0106	0.6244 ± 0.0238	0.6817 ± 0.0546
	F1	0.5688 ± 0.0055	0.4974 ± 0.0193	0.7553 ± 0.1598	0.6934 ± 0.0669	0.8298 ± 0.0000	0.4592 ± 0.0483	0.5461 ± 0.0071	0.7922 ± 0.0760	0.5347 ± 0.0192	0.5944 ± 0.0367
	NMI	0.0389 ± 0.0016	0.0012 ± 0.0015	0.0106 ± 0.0023	0.0465 ± 0.0073	0.0204 ± 0.0000	0.0016 ± 0.0020	0.0344 ± 0.0041	0.0300 ± 0.0207	0.0070 ± 0.0077	0.0382 ± 0.0220
	ARI	0.0258 ± 0.0062	0.0030 ± 0.0062	-0.0029 ± 0.0061	0.0966 ± 0.0306	-0.0144 ± 0.0000	0.0018 ± 0.0101	0.0875 ± 0.0071	0.0220 ± 0.0714	0.0243 ± 0.0111	0.0981 ± 0.0485
InsectWingbeat	ACC	0.1651 ± 0.0020	0.1003 ± 0.0000	0.1251 ± 0.0049	0.1099 ± 0.0024	0.1533 ± 0.0004	OOM	0.1451 ± 0.0010	0.1000 ± 0.0000	0.0872 ± 0.0071	0.1336 ± 0.0017
	F1	0.1623 ± 0.0020	0.0189 ± 0.0001	0.0945 ± 0.0222	0.0962 ± 0.0074	0.1062 ± 0.0051	OOM	0.1374 ± 0.0049	0.1818 ± 0.0000	0.0812 ± 0.0055	0.1209 ± 0.0043
	NMI	0.0428 ± 0.0011	0.0010 ± 0.0001	0.0090 ± 0.0015	0.1883 ± 0.0060	0.0339 ± 0.0008	OOM	0.0210 ± 0.0008	0.0000 ± 0.0000	0.0032 ± 0.0061	0.0146 ± 0.0015
	ARI	0.0216 ± 0.0005	0.0000 ± 0.0000	0.0027 ± 0.0013	0.0377 ± 0.0055	0.0130 ± 0.0007	OOM	0.0087 ± 0.0002	0.0000 ± 0.0000	0.0043 ± 0.0000	0.0059 ± 0.0006
JapaneseVowels	ACC	0.6286 ± 0.0530	0.1844 ± 0.0000	0.3203 ± 0.0273	0.4766 ± 0.0639	0.5198 ± 0.0223	0.1844 ± 0.0000	0.3369 ± 0.0199	0.1844 ± 0.0000	0.2619 ± 0.0271	0.8525 ± 0.0431
	F1	0.6617 ± 0.0303	0.0346 ± 0.0000	0.3325 ± 0.0365	0.4274 ± 0.0355	0.5551 ± 0.0053	0.0346 ± 0.0000	0.3069 ± 0.0236	0.3113 ± 0.0000	0.2755 ± 0.0282	0.8494 ± 0.0383
	NMI	0.5950 ± 0.0134	0.0000 ± 0.0000	0.2410 ± 0.0418	0.5250 ± 0.0397	0.4727 ± 0.0084	0.0000 ± 0.0000	0.2820 ± 0.0112	0.0000 ± 0.0000	0.1643 ± 0.0209	0.8113 ± 0.0180
	ARI	0.4771 ± 0.0389	0.0000 ± 0.0000	0.1269 ± 0.0412	0.3167 ± 0.0466	0.3049 ± 0.0281	0.0000 ± 0.0000	0.1511 ± 0.0210	0.0000 ± 0.0000	0.0648 ± 0.0171	0.7578 ± 0.0423
Libras	ACC	0.3822 ± 0.0052	0.3894 ± 0.0321	0.4011 ± 0.0315	0.3544 ± 0.0099	0.4317 ± 0.0256	0.1456 ± 0.0416	0.3844 ± 0.0208	0.2700 ± 0.0148	0.2839 ± 0.0116	0.3711 ± 0.0125
	F1	0.3931 ± 0.0193	0.3772 ± 0.0369	0.4028 ± 0.0406	0.2626 ± 0.0226	0.4438 ± 0.0298	0.0627 ± 0.0302	0.3700 ± 0.0242	0.2767 ± 0.0166	0.2853 ± 0.0157	0.3655 ± 0.0158
	NMI	0.5311 ± 0.0095	0.4697 ± 0.0327	0.4890 ± 0.0322	0.4698 ± 0.0245	0.5056 ± 0.0189	0.1311 ± 0.0783	0.5126 ± 0.0062	0.3101 ± 0.0185	0.3562 ± 0.0152	0.4806 ± 0.0259
	ARI	0.2484 ± 0.0082	0.2143 ± 0.0276	0.1881 ± 0.0295	0.2019 ± 0.0228	0.2423 ± 0.0142	0.0370 ± 0.0255	0.2532 ± 0.0127	0.0901 ± 0.0166	0.1246 ± 0.0146	0.2089 ± 0.0196
LSST	ACC	0.3434 ± 0.0022	0.1224 ± 0.0038	0.3672 ± 0.0074	0.2491 ± 0.0169	0.3327 ± 0.0037	0.3227 ± 0.0092	0.1578 ± 0.0113	0.3293 ± 0.0025	0.1805 ± 0.0098	0.2109 ± 0.0029
	F1	0.1918 ± 0.0134	0.0984 ± 0.0072	0.2055 ± 0.0297	0.2114 ± 0.0148	0.1689 ± 0.0141	0.0521 ± 0.0222	0.1149 ± 0.0053	0.2387 ± 0.0157	0.1376 ± 0.0084	0.1840 ± 0.0147
	NMI	0.1188 ± 0.0020	0.0311 ± 0.0056	0.1959 ± 0.0261	0.2066 ± 0.0198	0.1338 ± 0.0040	0.0237 ± 0.0303	0.0764 ± 0.0061	0.1398 ± 0.0038	0.0543 ± 0.0132	0.1396 ± 0.0043
	ARI	0.0453 ± 0.0007	0.0096 ± 0.0022	0.0936 ± 0.0182	0.0634 ± 0.0311	0.0859 ± 0.0106	0.0070 ± 0.0089	0.0222 ± 0.0040	0.0254 ± 0.0061	0.0207 ± 0.0068	0.0549 ± 0.0028
MotorImagery	ACC	0.5016 ± 0.0013	0.5360 ± 0.0147	0.5840 ± 0.0258	0.5048 ± 0.0072	0.5058 ± 0.0035	0.5111 ± 0.0059	0.5159 ± 0.0111	0.5127 ± 0.0048	0.5333 ± 0.0173	0.5508 ± 0.0070
	F1	0.3512 ± 0.0022	0.5084 ± 0.0319	0.5308 ± 0.0377	0.6586 ± 0.0706	0.3794 ± 0.0205	0.4565 ± 0.0416	0.5093 ± 0.0107	0.4401 ± 0.0338	0.5180 ± 0.0271	0.5362 ± 0.0175
	NMI	0.0003 ± 0.0002	0.0055 ± 0.0031	0.0540 ± 0.0291	0.0023 ± 0.0040	0.0011 ± 0.0008	0.0017 ± 0.0016	0.0012 ± 0.0011	0.0012 ± 0.0004	0.0044 ± 0.0042	0.0096 ± 0.0034
	ARI	-0.0002 ± 0.0000	0.0040 ± 0.0035	0.0254 ± 0.0160	-0.0003 ± 0.0004	-0.0003 ± 0.0001	-0.0010 ± 0.0011	-0.0010 ± 0.0014	-0.0006 ± 0.0001	0.0033 ± 0.0059	0.0082 ± 0.0028
NATOPS	ACC	0.6537 ± 0.0975	0.4861 ± 0.0512	0.3978 ± 0.0514	0.5672 ± 0.0859	0.4350 ± 0.0268	0.3267 ± 0.0919	0.3817 ± 0.0194	0.3522 ± 0.0811	0.2972 ± 0.0249	0.5022 ± 0.0157
	F1	0.6529 ± 0.0968	0.4643 ± 0.0539	0.4342 ± 0.1010	0.5167 ± 0.0816	0.4372 ± 0.0599	0.2052 ± 0.0868	0.3871 ± 0.0168	0.2131 ± 0.0691	0.2951 ± 0.0249	0.4958 ± 0.0103
	NMI	0.6454 ± 0.0805	0.4355 ± 0.0551	0.2990 ± 0.1190	0.5575 ± 0.0724	0.3896 ± 0.0459	0.2876 ± 0.1673	0.1897 ± 0.0207	0.1515 ± 0.1613	0.1097 ± 0.0324	0.5159 ± 0.0230
	ARI	0.5091 ± 0.1162	0.3056 ± 0.0590	0.1759 ± 0.0826	0.4179 ± 0.0983	0.2563 ± 0.0459	0.1603 ± 0.1191	0.1147 ± 10.0214	0.2551 ± 0.1261	0.0549 ± 0.0222	0.3489 ± 0.0284
PEMS-SF	ACC	0.4864 ± 0.0558	0.1968 ± 0.0065	0.4301 ± 0.0842	0.2268 ± 0.0179	0.3582 ± 0.0366	0.2514 ± 0.0897	0.4268 ± 0.0172	0.2377 ± 0.0102	0.4409 ± 0.0466	0.4436 ± 0.0844
	F1	0.5110 ± 0.0447	0.1941 ± 0.0064	0.4202 ± 0.0807	0.2263 ± 0.0204	0.3782 ± 0.0333	0.1185 ± 0.0734	0.4239 ± 0.0214	0.2370 ± 0.0154	0.4683 ± 0.0591	0.4902 ± 0.1198
	NMI	0.4510 ± 0.0416	0.0348 ± 0.0119	0.3687 ± 0.1015	0.1270 ± 0.0168	0.3336 ± 0.0281	0.2715 ± 0.2226	0.4593 ± 0.0196	0.1742 ± 0.0226	0.4167 ± 0.0441	0.4570 ± 0.1104
	ARI	0.2911 ± 0.0478	0.0030 ± 0.0034	0.2134 ± 0.0924	0.0374 ± 0.0156	0.1673 ± 0.0312	0.1333 ± 0.1105	0.2557 ± 0.0121	0.0651 ± 0.0082	0.2471 ± 0.0345	0.2839 ± 0.0947
PenDigits	ACC	0.4246 ± 0.0409	0.4128 ± 0.0672	0.4089 ± 0.0499	0.6548 ± 0.0159	0.3484 ± 0.0354	0.2670 ± 0.0505	0.4078 ± 0.0162	0.3647 ± 0.0150	0.4687 ± 0.0517	0.6841 ± 0.0284
	F1	0.4116 ± 0.0303	0.3971 ± 0.0746	0.4086 ± 0.0578	0.5693 ± 0.0171	0.3548 ± 0.0321	0.1710 ± 0.0688	0.3825 ± 0.0213	0.3747 ± 0.0150	0.4542 ± 0.0512	0.6938 ± 0.0411
	NMI	0.3233 ± 0.0199	0.3405 ± 0.0616	0.3994 ± 0.0617	0.6583 ± 0.0086	0.3059 ± 0.0165	0.2076 ± 0.0559	0.3390 ± 0.0131	0.2611 ± 0.0165	0.4348 ± 0.0630	0.6485 ± 0.0189
	ARI	0.2218 ± 0.0289	0.2151 ± 0.0560	0.2313 ± 0.0515	0.5196 ± 0.0193	0.1503 ± 0.0122	0.1033 ± 0.0425	0.2183 ± 0.0158	0.1528 ± 0.0112	0.2822 ± 0.0658	0.5305 ± 0.0276
PhonemeSpectra	ACC	0.1149 ± 0.0003	0.0633 ± 0.0006	0.0963 ± 0.0032	0.1124 ± 0.0038	0.1283 ± 0.0024	0.0454 ± 0.0059	0.0916 ± 0.0028	0.0871 ± 0.0023	0.0812 ± 0.0071	0.1127 ± 0.0015
	F1	0.1159 ± 0.0004	0.0605 ± 0.0010	0.0944 ± 0.0004	0.1048 ± 0.0067	0.1242 ± 0.0038	0.0099 ± 0.0028	0.0913 ± 0.0033	0.0845 ± 0.0017	0.0786 ± 0.0080	0.1126 ± 0.0017
	NMI	0.1793 ± 0.0023	0.0451 ± 0.0015	0.1386 ± 0.0089	0.2039 ± 0.0025	0.2085 ± 0.0038	0.0366 ± 0.0136	0.1090 ± 0.0036	0.1174 ± 0.0070	0.0932 ± 0.0191	0.1758 ± 0.0026
	ARI	0.0339 ± 0.0007	0.0038 ± 0.0003	0.0166 ± 0.0040	0.0424 ± 0.0048	0.0449 ± 0.0011	0.0046 ± 0.0026	0.0178 ± 0.0015	0.0186 ± 0.0020	0.0143 ± 0.0045	0.0334 ± 0.0008
RocketSports	ACC	0.4172 ± 0.0058	0.3921 ± 0.0150	0.4355 ± 0.0286	0.6205 ± 0.0400	0.4257 ± 0.0251	0.3129 ± 0.0296	0.3551 ± 0.0324	0.5036 ± 0.0475	0.3743 ± 0.0375	0.6007 ± 0.0721
	F1	0.4138 ± 0.0065	0.3898 ± 0.0135	0.4064 ± 0.0374	0.5330 ± 0.0327	0.4067 ± 0.0328	0.2105 ± 0.0508	0.3529 ± 0.0340	0.5273 ± 0.0345	0.3667 ± 0.0383	0.6006 ± 0.0707
	NMI	0.1398 ± 0.0042	0.1024 ± 0.0386	0.1138 ± 0.0264	0.4687 ± 0.0339	0.1250 ± 0.0289	0.0688 ± 0.0371	0.0601 ± 0.0348	0.3279 ± 0.0420	0.0818 ± 0.0394	0.3488 ± 0.0845
	ARI	0.0880 ± 0.0042	0.0590 ± 0.0167	0.0776 ± 0.0307	0.3723 ± 0.0445	0.0924 ± 0.0200	0.0127 ± 0.0191	0.0400 ± 0.0293	0.2394 ± 0.0441	0.0488 ± 0.0282	0.2858 ± 0.0965
SelfRegulationSCP1	ACC	0.5362 ± 0.0031	0.5447 ± 0.0307	0.7126 ± 0.0431	0.5027 ± 0.0000	0.6096 ± 0.0472	0.5537 ± 0.0342	0.5586 ± 0.0087	0.7340 ± 0.0059	0.5811 ± 0.0402	0.7754 ± 0.0128
	F1	0.5081 ± 0.0040	0.5434 ± 0.0324	0.7073 ± 0.0419	0.7065 ± 0.0000	0.6082 ± 0.0487	0.4723 ± 0.0964	0.5584 ± 0.0087	0.7184 ± 0.0076	0.5676 ± 0.0372	0.7745 ± 0.0125
	NMI	0.0059 ± 0.0009	0.0086 ± 0.0093	0.1642 ± 0.0748	0.0000 ± 0.0000	0.0421 ± 0.0290	0.0261 ± 0.0139	0.0102 ± 0.0031	0.2493 ± 0.0044	0.0328 ± 0.0305	0.2382 ± 0.0246
	ARI	0.0039 ± 0.0009	0.0100 ± 0.0128	0.1857 ± 0.0753	0.0000 ± 0.0000	0.0553 ± 0.0393	0.0154 ± 0.0124	0.0123 ± 0.0043	0.2182 ± 0.0111	0.0313 ± 0.0296	0.3028 ± 0.0274
SelfRegulationSCP2	ACC	0.5163 ± 0.0051	0.5258 ± 0.0162	0.5811 ± 0.0143	0.5027 ± 0.0000	0.5521 ± 0.0096	0.5347 ± 0.0210	0.5053 ± 0.0069	0.5432 ± 0.0054	0.5811 ± 0.0402	0.5253 ± 0.0141
	F1	0.4913 ± 0.0063	0.5242 ± 0.0166	0.5657 ± 0.0301	0.7065 ± 0.0000	0.5448 ± 0.0077	0.5302 ± 0.0239	0.5039 ± 0.0073	0.5113 ± 0.0110	0.5676 ± 0.0372	0.5163 ± 0.0084
	NMI	0.0011 ± 0.0006	0.0027 ± 0.0027	0.0234 ± 0.0043	0.0000 ± 0.0000	0.0090 ± 0.0034	0.0049 ± 0.0048	0.0002 ± 0.0004	0.0083 ± 0.0020	0.0328 ± 0.0305	0.0034 ± 0.0044
	ARI	-0.0010 ± 0.0006	0.0011 ± 0.0036	0.0224 ± 0.0083	0.0000 ± 0.0000	0.0088 ± 0.0040	0.0041 ± 0.0065	-0.0023 ± 0.0005	0.0056 ± 0.0018	0.0313 ± 0.0296	0.0009 ± 0.0038
SpokenArabicDigits	ACC	0.4902 ± 0.0258	0.1000 ± 0.0000	0.2415 ± 0.0313	0.3869 ± 0.0543	0.4655 ± 0.1073	0.1889 ± 0.0452	0.3692 ± 0.0053	0.1000 ± 0.0000	0.3054 ± 0.0040	0.4420 ± 0.0257
	F1	0.4939 ± 0.0288	0.0182 ± 0.0000	0.2320 ± 0.0343	0.3303 ± 0.0362	0.4617 ± 0.1044	0.1196 ± 0.0519	0.3713 ± 0.0084	0.1819 ± 0.0000	0.3030 ± 0.0060	0.4460 ± 0.0206
	NMI	0.4740 ± 0.0129	0.0000 ± 0.0000	0.1566 ± 0.0474	0.4331 ± 0						

TABLE III
ABLATION STUDIES ON MODULE DESIGNS OF SIGMA.

Datasets (1-15)	Components		Metrics			
	CCMA	MEVG	ACC	F1	NMI	ARI
ArticulatoryWordRecognition	✓	✓	0.7329 ± 0.0395	0.7696 ± 0.0443	0.8037 ± 0.0246	0.6029 ± 0.0470
		✓	0.7155 ± 0.0137	0.7451 ± 0.0215	0.7824 ± 0.0132	0.5728 ± 0.0248
	✓		0.7739 ± 0.0437	0.7997 ± 0.0481	0.8255 ± 0.0189	0.6570 ± 0.0427
		✓	0.6918 ± 0.0306	0.7336 ± 0.0245	0.7709 ± 0.0187	0.5517 ± 0.0369
AtrialFibrillation	✓	✓	0.4933 ± 0.0327	0.4720 ± 0.0232	0.1215 ± 0.0474	0.0443 ± 0.0492
		✓	0.4867 ± 0.0163	0.4331 ± 0.0116	0.1050 ± 0.0477	0.0404 ± 0.0051
	✓		0.4800 ± 0.0400	0.4681 ± 0.0854	0.1365 ± 0.0448	0.0664 ± 0.0421
		✓	0.4600 ± 0.0327	0.5024 ± 0.0485	0.1000 ± 0.0087	0.0346 ± 0.0071
BasicMotions	✓	✓	0.9675 ± 0.0150	0.9671 ± 0.0154	0.9273 ± 0.0277	0.9169 ± 0.0363
		✓	0.9825 ± 0.0127	0.9824 ± 0.0129	0.9497 ± 0.0366	0.9533 ± 0.0338
	✓		0.8900 ± 0.0137	0.9203 ± 0.0746	0.8924 ± 0.0754	0.8361 ± 0.1360
		✓	0.9700 ± 0.0100	0.9698 ± 0.0102	0.9138 ± 0.0287	0.9202 ± 0.0263
CharacterTrajectories	✓	✓	0.7879 ± 0.0499	0.8091 ± 0.0408	0.8463 ± 0.0284	0.7214 ± 0.0576
		✓	0.7824 ± 0.0547	0.8177 ± 0.0436	0.8416 ± 0.0367	0.7078 ± 0.0644
	✓		0.7371 ± 0.0227	0.7610 ± 0.0346	0.8038 ± 0.0198	0.6417 ± 0.0303
		✓	0.7369 ± 0.0590	0.7699 ± 0.0708	0.8138 ± 0.0468	0.6556 ± 0.0750
Cricket	✓	✓	0.8244 ± 0.0255	0.8463 ± 0.0115	0.8522 ± 0.0167	0.7259 ± 0.0299
		✓	0.8489 ± 0.0349	0.8739 ± 0.0420	0.8737 ± 0.0444	0.7589 ± 0.0659
	✓		0.7811 ± 0.0658	0.8191 ± 0.0588	0.8346 ± 0.0364	0.6817 ± 0.0702
		✓	0.8078 ± 0.0114	0.8292 ± 0.0241	0.8376 ± 0.0130	0.6937 ± 0.0195
DuckDuckGeese	✓	✓	0.3900 ± 0.0141	0.3730 ± 0.0193	0.1749 ± 0.0375	0.0795 ± 0.0345
		✓	0.3700 ± 0.0179	0.3600 ± 0.0253	0.1143 ± 0.0234	0.0526 ± 0.0108
	✓		0.3280 ± 0.0421	0.2966 ± 0.0425	0.0869 ± 0.0246	0.0196 ± 0.0205
		✓	0.3680 ± 0.0194	0.3449 ± 0.0235	0.1229 ± 0.0145	0.0306 ± 0.0112
ERing	✓	✓	0.7307 ± 0.0629	0.7287 ± 0.0680	0.5996 ± 0.0590	0.5124 ± 0.0760
		✓	0.6867 ± 0.0669	0.7380 ± 0.0341	0.6238 ± 0.0426	0.5008 ± 0.0669
	✓		0.7300 ± 0.0485	0.7313 ± 0.0475	0.6198 ± 0.0400	0.5226 ± 0.0569
		✓	0.7173 ± 0.0259	0.7082 ± 0.0253	0.6112 ± 0.0390	0.5033 ± 0.0415
EigenWorms	✓	✓	0.4193 ± 0.0788	0.3646 ± 0.0438	0.1612 ± 0.0594	0.1515 ± 0.0726
		✓	0.4131 ± 0.0101	0.3546 ± 0.0347	0.2196 ± 0.0211	0.1907 ± 0.0156
	✓		0.4425 ± 0.0451	0.4077 ± 0.0359	0.1829 ± 0.0256	0.1469 ± 0.0344
		✓	0.3822 ± 0.0462	0.3397 ± 0.0392	0.1713 ± 0.0600	0.1368 ± 0.0716
Epilepsy	✓	✓	0.4655 ± 0.0229	0.4573 ± 0.0233	0.2355 ± 0.0362	0.1668 ± 0.0336
		✓	0.4640 ± 0.0291	0.4648 ± 0.0300	0.2217 ± 0.0332	0.1511 ± 0.0259
	✓		0.4720 ± 0.0368	0.4707 ± 0.0382	0.2330 ± 0.0451	0.1551 ± 0.0279
		✓	0.4829 ± 0.0238	0.4847 ± 0.0243	0.2418 ± 0.0298	0.1637 ± 0.0212
EthanolConcentration	✓	✓	0.3061 ± 0.0208	0.2766 ± 0.0066	0.0199 ± 0.0156	0.0108 ± 0.0123
		✓	0.3015 ± 0.0099	0.2777 ± 0.0082	0.0114 ± 0.0033	0.0049 ± 0.0030
	✓		0.3088 ± 0.0183	0.2881 ± 0.0252	0.0184 ± 0.0066	0.0104 ± 0.0062
		✓	0.2950 ± 0.0092	0.2797 ± 0.0108	0.0124 ± 0.0061	0.0045 ± 0.0039
FaceDetection	✓	✓	0.5107 ± 0.0042	0.5100 ± 0.0049	0.0004 ± 0.0003	0.0004 ± 0.0004
		✓	0.5051 ± 0.0028	0.5050 ± 0.0028	0.0001 ± 0.0001	0.0000 ± 0.0001
	✓		0.5101 ± 0.0023	0.5092 ± 0.0027	0.0003 ± 0.0001	0.0003 ± 0.0002
		✓	0.5063 ± 0.0076	0.5061 ± 0.0077	0.0003 ± 0.0005	0.0003 ± 0.0007
FingerMovements	✓	✓	0.5188 ± 0.0085	0.5142 ± 0.0086	0.0013 ± 0.0007	0.0006 ± 0.0010
		✓	0.5168 ± 0.0101	0.5123 ± 0.0056	0.0013 ± 0.0015	0.0008 ± 0.0017
	✓		0.5168 ± 0.0128	0.5125 ± 0.0160	0.0013 ± 0.0014	0.0005 ± 0.0019
		✓	0.5130 ± 0.0074	0.5127 ± 0.0073	0.0006 ± 0.0007	0.0015 ± 0.0010
HandMovementDirection	✓	✓	0.3333 ± 0.0143	0.3298 ± 0.0171	0.0205 ± 0.0063	0.0060 ± 0.0052
		✓	0.3051 ± 0.0154	0.3027 ± 0.0162	0.0139 ± 0.0036	0.0006 ± 0.0029
	✓		0.3179 ± 0.0249	0.3148 ± 0.0246	0.0201 ± 0.0119	0.0049 ± 0.0111
		✓	0.3060 ± 0.0126	0.3034 ± 0.0116	0.0144 ± 0.0029	0.0011 ± 0.0025
Handwriting	✓	✓	0.2386 ± 0.0181	0.2433 ± 0.0195	0.3461 ± 0.0286	0.1006 ± 0.0180
		✓	0.2102 ± 0.0132	0.2103 ± 0.0137	0.3030 ± 0.0188	0.0770 ± 0.0111
	✓		0.2444 ± 0.0113	0.2460 ± 0.0156	0.3503 ± 0.0160	0.1050 ± 0.0109
		✓	0.2224 ± 0.0130	0.2234 ± 0.0113	0.3212 ± 0.0209	0.0852 ± 0.0100
Heartbeat	✓	✓	0.6636 ± 0.0589	0.5679 ± 0.0507	0.0391 ± 0.0232	0.0689 ± 0.0404
		✓	0.6826 ± 0.0369	0.6002 ± 0.0271	0.0438 ± 0.0108	0.0952 ± 0.0277
	✓		0.6841 ± 0.0410	0.5474 ± 0.0747	0.0328 ± 0.0106	0.0620 ± 0.0591
		✓	0.7066 ± 0.0091	0.6463 ± 0.0947	0.0365 ± 0.0108	0.0846 ± 0.0516

lations on all 30 datasets. The complete model generally gives stronger or competitive results, and the joint removal of CCMA and MEVG leads to the clearest degradation on datasets such as ArticulatoryWordRecognition, CharacterTrajectories, DuckDuckGeese, PEMS-SF, PenDigits, PhonemeSpectra, and UWaveGestureLibrary. Removing CCMA weakens the admission decision before encoder-side interaction, so background-dominated patches can still shape the representation through frequent feature interactions. Removing MEVG weakens the organization of partially observed content, so the retained patches are more likely to be compressed into a narrow representation basis. Some individual datasets favor one ablated variant on a single metric, which is expected in unsupervised MTS clustering because different datasets emphasize different similarity structures. However, the complete table shows a stable pattern. CCMA provides source-level pruning, MEVG serves as semantic compensation after partial observation, and their joint design reduces the risk that pruning becomes either overly aggressive or semantically narrow.

TABLEs V and VI report the complete route-level ablations.

TABLE IV
ABLATION STUDIES ON MODULE DESIGNS OF SIGMA.

Datasets (16-30)	Components		Metrics			
	CCMA	MEVG	ACC	F1	NMI	ARI
InsectWingbeat	✓	✓	0.1344 ± 0.0023	0.1202 ± 0.0033	0.0153 ± 0.0012	0.0061 ± 0.0007
		✓	0.1351 ± 0.0015	0.1206 ± 0.0036	0.0147 ± 0.0008	0.0064 ± 0.0001
	✓		0.1304 ± 0.0032	0.1115 ± 0.0076	0.0132 ± 0.0011	0.0048 ± 0.0013
		✓	0.1323 ± 0.0039	0.1099 ± 0.0069	0.0139 ± 0.0017	0.0050 ± 0.0011
JapaneseVowels	✓	✓	0.8647 ± 0.0447	0.8626 ± 0.0432	0.8195 ± 0.0260	0.7682 ± 0.0565
		✓	0.8162 ± 0.0393	0.8557 ± 0.0416	0.8044 ± 0.0126	0.7321 ± 0.0429
	✓		0.8750 ± 0.0347	0.8840 ± 0.0294	0.8220 ± 0.0239	0.7847 ± 0.0385
		✓	0.8800 ± 0.0306	0.8802 ± 0.0270	0.8140 ± 0.0272	0.7648 ± 0.0553
LSST	✓	✓	0.2042 ± 0.0087	0.1805 ± 0.0063	0.1456 ± 0.0063	0.0555 ± 0.0044
		✓	0.2018 ± 0.0106	0.1853 ± 0.0156	0.1475 ± 0.0125	0.0561 ± 0.0074
	✓		0.2096 ± 0.0131	0.1792 ± 0.0080	0.1458 ± 0.0102	0.0567 ± 0.0093
		✓	0.2048 ± 0.0134	0.1824 ± 0.0031	0.1553 ± 0.0062	0.0589 ± 0.0052
Libras	✓	✓	0.3156 ± 0.0374	0.3162 ± 0.0405	0.4101 ± 0.0302	0.1502 ± 0.0253
		✓	0.2839 ± 0.0032	0.2979 ± 0.0079	0.3439 ± 0.0107	0.1012 ± 0.0057
	✓		0.3272 ± 0.0278	0.3460 ± 0.0497	0.4212 ± 0.0300	0.1600 ± 0.0271
		✓	0.2883 ± 0.0069	0.2959 ± 0.0148	0.3481 ± 0.0125	0.1084 ± 0.0080
MotorImagery	✓	✓	0.5323 ± 0.0147	0.5271 ± 0.0154	0.0039 ± 0.0034	0.0025 ± 0.0046
		✓	0.5360 ± 0.0178	0.5196 ± 0.0260	0.0053 ± 0.0041	0.0041 ± 0.0050
	✓		0.5307 ± 0.0224	0.5249 ± 0.0284	0.0042 ± 0.0048	0.0032 ± 0.0066
		✓	0.5238 ± 0.0115	0.4965 ± 0.0034	0.0035 ± 0.0038	0.0007 ± 0.0027
NATOPS	✓	✓	0.4756 ± 0.0180	0.4638 ± 0.0129	0.4697 ± 0.0383	0.3103 ± 0.0385
		✓	0.4533 ± 0.0457	0.4550 ± 0.0494	0.4481 ± 0.0457	0.2797 ± 0.0460
	✓		0.4528 ± 0.0511	0.4554 ± 0.0583	0.4444 ± 0.0859	0.2844 ± 0.0753
		✓	0.4750 ± 0.0072	0.4797 ± 0.0059	0.4753 ± 0.0118	0.3082 ± 0.0133
PEMS-SF	✓	✓	0.3905 ± 0.0724	0.4364 ± 0.0969	0.3625 ± 0.1185	0.2028 ± 0.0839
		✓	0.2768 ± 0.0192	0.2752 ± 0.0175	0.2469 ± 0.0372	0.1170 ± 0.0271
	✓		0.4159 ± 0.0708	0.4722 ± 0.0819	0.4329 ± 0.0928	0.2465 ± 0.0908
		✓	0.2550 ± 0.0164	0.2553 ± 0.0131	0.2111 ± 0.0293	0.0900 ± 0.0244
PenDigits	✓	✓	0.6796 ± 0.0497	0.6724 ± 0.0488	0.6440 ± 0.0381	0.5249 ± 0.0488
		✓	0.5957 ± 0.0208	0.5949 ± 0.0207	0.5736 ± 0.0164	0.4065 ± 0.0264
	✓		0.6614 ± 0.0378	0.6457 ± 0.0436	0.6188 ± 0.0319	0.4953 ± 0.0431
		✓	0.5954 ± 0.0262	0.5955 ± 0.0250	0.5702 ± 0.0197	0.4034 ± 0.0284
PhonemeSpectra	✓	✓	0.1125 ± 0.0040	0.1119 ± 0.0039	0.1765 ± 0.0084	0.0333 ± 0.0020
		✓	0.1036 ± 0.0047	0.1044 ± 0.0044	0.1576 ± 0.0107	0.0277 ± 0.0029
	✓		0.1036 ± 0.0016	0.1027 ± 0.0021	0.1568 ± 0.0039	0.0289 ± 0.0014
		✓	0.1036 ± 0.0016	0.1027 ± 0.0021	0.1568 ± 0.0039	0.0289 ± 0.0014
RacketSports	✓	✓	0.5908 ± 0.0704	0.5918 ± 0.0700	0.3431 ± 0.0775	0.2754 ± 0.0886
		✓	0.6330 ± 0.0454	0.6263 ± 0.0497	0.4037 ± 0.0637	0.3443 ± 0.0751
	✓		0.5340 ± 0.0979	0.5410 ± 0.0966	0.3100 ± 0.1019	0.2449 ± 0.1091
		✓	0.6026 ± 0.0949	0.6012 ± 0.0964	0.3912 ± 0.1121	0.3290 ± 0.1141
SelfRegulation SCP1	✓	✓	0.7747 ± 0.0140	0.7738 ± 0.0136	0.2371 ± 0.0269	0.3014 ± 0.0296
		✓	0.7647 ± 0.0254	0.7635 ± 0.0258	0.2224 ± 0.0410	0.2816 ± 0.0510
	✓		0.7740 ± 0.0170	0.7730 ± 0.0160	0.2387 ± 0.0412	0.3002 ± 0.0376
		✓	0.7647 ± 0.0103	0.7644 ± 0.0099	0.2150 ± 0.0213	0.2794 ± 0.0224
SelfRegulation SCP2	✓	✓	0.5253 ± 0.0151	0.5162 ± 0.0079	0.0036 ± 0.0050	0.0010 ± 0.0042
		✓	0.5279 ± 0.0107	0.5270 ± 0.0093	0.0026 ± 0.0022	0.0010 ± 0.0029
	✓		0.5147 ± 0.0106	0.5141 ± 0.0105	0.0010 ± 0.0013	0.0013 ± 0.0017
		✓	0.5142 ± 0.0032	0.5141 ± 0.0033	0.0006 ± 0.0003	0.0018 ± 0.0004
Spoken ArabicDigits	✓	✓	0.4260 ± 0.0202	0.4328 ± 0.0168	0.3868 ± 0.0165	0.2365 ± 0.0194
		✓	0.4712 ± 0.0196	0.4778 ± 0.0219	0.4407 ± 0.0153	0.2844 ± 0.0124
	✓		0.4477 ± 0.0191	0.4502 ± 0.0231	0.4457 ± 0.0383	0.2802 ± 0.0395
		✓	0.4637 ± 0.0203	0.4691 ± 0.0204	0.4426 ± 0.0119	0.2821 ± 0.0086
StandWalkJump	✓	✓	0.4000 ± 0.0637	0.3740 ± 0.0724	0.0441 ± 0.0480	0.0171 ± 0.0522
		✓	0.4000 ± 0.0277	0.3749 ± 0.0372	0.0458 ± 0.0276	0.0173 ± 0.0277
	✓		0.4222 ± 0.0502	0.3944 ± 0.0506	0.0417 ± 0.0306	0.0215 ± 0.0349
		✓	0.4148 ± 0.0544	0.3852 ± 0.0626	0.0547 ± 0.0419	0.0341 ± 0.0161
UWaveGesture Library	✓	✓	0.4164 ± 0.0435	0.4043 ± 0.0445	0.3456 ± 0.0479	0.1996 ± 0.0452
		✓	0.3373 ± 0.0086	0.3156 ± 0.0090	0.1918 ± 0.0200	0.0954 ± 0.0114
	✓		0.3945 ± 0.0411	0.3881 ± 0.0581	0.2768 ± 0.0378	0.1487 ± 0.0261
		✓	0.3127 ± 0.0232	0.2967 ± 0.0177	0.1782 ± 0.0133	0.0860 ± 0.0156

TABLE V
ABLATION STUDIES ON PATH OBJECTIVES OF SIGMA (PART 1/2).

Datasets (1-15)	Components		Metrics			
	Recon	Contra	ACC	F1	NMI	ARI
ArticulatoryWord Recognition	✓	✓	0.7329 ± 0.0395	0.7696 ± 0.0443	0.8037 ± 0.0246	0.6029 ± 0.0470
	✓	✓	0.6035 ± 0.0317	0.6212 ± 0.0261	0.7016 ± 0.0278	0.4614 ± 0.0399
	✓	✓	0.8247 ± 0.0562	0.8514 ± 0.0558	0.8747 ± 0.0426	0.7445 ± 0.0793
AtrialFibrillation	✓	✓	0.4933 ± 0.0327	0.4720 ± 0.0232	0.1215 ± 0.0474	0.0443 ± 0.0492
	✓	✓	0.4933 ± 0.0249	0.4763 ± 0.0291	0.0833 ± 0.0242	0.0169 ± 0.0203
	✓	✓	0.4733 ± 0.0389	0.4520 ± 0.0542	0.0765 ± 0.0312	0.0190 ± 0.0333
BasicMotions	✓	✓	0.9675 ± 0.0150	0.9671 ± 0.0154	0.9273 ± 0.0277	0.9169 ± 0.0363
	✓	✓	0.9050 ± 0.1226	0.9319 ± 0.0784	0.9161 ± 0.0911	0.8633 ± 0.1575
	✓	✓	0.6600 ± 0.0366	0.8105 ± 0.0355	0.7754 ± 0.0499	0.6063 ± 0.0482
Character Trajectories	✓	✓	0.7879 ± 0.0499	0.8091 ± 0.0408	0.8463 ± 0.0284	0.7214 ± 0.0576
	✓	✓	0.6714 ± 0.0832	0.7199 ± 0.0942	0.7443 ± 0.0645	0.5830 ± 0.1023
	✓	✓	0.7389 ± 0.0708	0.7674 ± 0.0674	0.8189 ± 0.0503	0.6750 ± 0.0821
Cricket	✓	✓	0.651 ± 0.0000	0.1222 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
	✓	✓	0.8244 ± 0.0255	0.8463 ± 0.0115	0.8522 ± 0.0167	0.7259 ± 0.0299
	✓	✓	0.7900 ± 0.1095	0.8001 ± 0.1052	0.8279 ± 0.0699	0.6912 ± 0.1158
DuckDuckGeese	✓	✓	0.7511 ± 0.0708	0.7828 ± 0.0339	0.7682 ± 0.0476	0.5964 ± 0.0836
	✓	✓	0.1000 ± 0.0333	0.2018 ± 0.0960	0.0721 ± 0.1442	0.0121 ± 0.0241
	✓	✓	0.3900 ± 0.0141	0.3730 ± 0.0193	0.1749 ± 0.0375	0.0795 ± 0.0345
ERing	✓	✓	0.4740 ± 0.0546	0.4690 ± 0.0553	0.2439 ± 0.0457	0.1618 ± 0.0523
	✓	✓	0.3880 ± 0.0306	0.3682 ± 0.0377	0.1562 ± 0.0375	0.0621 ± 0.0248
	✓	✓	0.2000 ± 0.0000	0.3333 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
EigenWorms	✓	✓	0.7307 ± 0.0629	0.7287 ± 0.0680	0.5996 ± 0.0590	0.5124 ± 0.0760
	✓	✓	0.7533 ± 0.0419	0.7553 ± 0.0421	0.6735 ± 0.0240	0.5854 ± 0.0353
	✓	✓	0.8020 ± 0.0937	0.8006 ± 0.0951	0.7408 ± 0.0702	0.6688 ± 0.1070
Epilepsy	✓	✓	0.1667 ± 0.0000	0.2857 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
	✓	✓	0.4193 ± 0.0788	0.3646 ± 0.0438	0.1612 ± 0.0594	0.1515 ± 0.0726
	✓	✓	0.4556 ± 0.0523	0.4150 ± 0.0632	0.2774 ± 0.0369	0.2479 ± 0.0381
Ethanol Concentration	✓	✓	0.3382 ± 0.0317	0.3009 ± 0.0439	0.0843 ± 0.0295	0.0512 ± 0.0158
	✓	✓	0.4247 ± 0.0000	0.5962 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
	✓	✓	0.4655 ± 0.0229	0.4573 ± 0.0233	0.2355 ± 0.0362	0.1668 ± 0.0336
FaceDetection	✓	✓	0.5716 ± 0.0429	0.5746 ± 0.0356	0.3908 ± 0.0509	0.3182 ± 0.0499
	✓	✓	0.4320 ± 0.0292	0.4259 ± 0.0276	0.2047 ± 0.0333	0.1442 ± 0.0279
	✓	✓	0.4291 ± 0.0332	0.4217 ± 0.0355	0.1670 ± 0.0517	0.1224 ± 0.0370
FingerMovements	✓	✓	0.3061 ± 0.0208	0.2766 ± 0.0066	0.0199 ± 0.0156	0.0108 ± 0.0123
	✓	✓	0.2916 ± 0.0080	0.2841 ± 0.0078	0.0129 ± 0.0041	0.0032 ± 0.0014
	✓	✓	0.3115 ± 0.0143	0.2799 ± 0.0252	0.0208 ± 0.0187	0.0144 ± 0.0106
HandMovement Direction	✓	✓	0.3179 ± 0.0130	0.3277 ± 0.0507	0.0442 ± 0.0192	0.0238 ± 0.0127
	✓	✓	0.5107 ± 0.0042	0.5100 ± 0.0049	0.0004 ± 0.0003	0.0004 ± 0.0004
	✓	✓	0.5090 ± 0.0055	0.5086 ± 0.0056	0.0003 ± 0.0003	0.0003 ± 0.0004
Handwriting	✓	✓	0.5900 ± 0.0000	0.6667 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
	✓	✓	0.5000 ± 0.0000	0.6667 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
	✓	✓	0.5188 ± 0.0085	0.5142 ± 0.0086	0.0013 ± 0.0007	0.0006 ± 0.0010
Heartbeat	✓	✓	0.5135 ± 0.0097	0.5114 ± 0.0111	0.0008 ± 0.0009	0.0013 ± 0.0012
	✓	✓	0.5231 ± 0.0102	0.5188 ± 0.0102	0.0019 ± 0.0012	0.0002 ± 0.0016
	✓	✓	0.5000 ± 0.0000	0.6667 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
RacketSports	✓	✓	0.3333 ± 0.0143	0.3298 ± 0.0171	0.0205 ± 0.0063	0.0060 ± 0.0052
	✓	✓	0.3077 ± 0.0162	0.3061 ± 0.0171	0.0148 ± 0.0062	0.0005 ± 0.0053
	✓	✓	0.3111 ± 0.0208	0.3077 ± 0.0221	0.0215 ± 0.0075	0.0053 ± 0.0072
Self RegulationSCP1	✓	✓	0.3068 ± 0.0068	0.2901 ± 0.0134	0.0201 ± 0.0043	0.0021 ± 0.0020
	✓	✓	0.2386 ± 0.0181	0.2433 ± 0.0195	0.3461 ± 0.0286	0.1006 ± 0.0180
	✓	✓	0.1706 ± 0.0061	0.1741 ± 0.0083	0.2227 ± 0.0045	0.0385 ± 0.0019
Self RegulationSCP2	✓	✓	0.2054 ± 0.0071	0.2071 ± 0.0059	0.2809 ± 0.0091	0.0663 ± 0.0054
	✓	✓	0.1428 ± 0.0060	0.1408 ± 0.0057	0.1813 ± 0.0062	0.0217 ± 0.0024
	✓	✓	0.6636 ± 0.0589	0.5679 ± 0.0507	0.0391 ± 0.0232	0.0689 ± 0.0404
StandWalkJump	✓	✓	0.5628 ± 0.0376	0.5053 ± 0.0802	0.0459 ± 0.0155	0.0065 ± 0.0412
	✓	✓	0.7002 ± 0.0176	0.7537 ± 0.1522	0.0170 ± 0.0068	0.0161 ± 0.0034
	✓	✓	0.7213 ± 0.0000	0.8381 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000

decides which patches can participate in feature interaction. MEVG organizes the admitted content into complementary endogenous views. The reconstruction route keeps the pruned representation semantically traceable, and the contrastive route converts the organized views into prototype-aware clusters. Consequently, SIGMA does not rely on a single isolated component. Admission, view organization, semantic restoration, and cluster refinement occupy distinct but connected roles, which disentangles redundancy suppression from semantic preservation and then integrates the retained semantics with cluster refinement in a compact prune-and-preserve pipeline.

C. Additional Training-Loss Curves

The main paper answers Q3 by examining whether source-level admission remains efficient and stable under different data scales, training stages, and admission settings. The loss-curve analysis in the main paper focuses on four representative datasets and checks whether cluster-oriented objectives can be introduced after semantic warm-up without causing unstable optimization. This subsection extends the

TABLE VI
ABLATION STUDIES ON PATH OBJECTIVES OF SIGMA (PART 2/2).

Datasets (16-30)	Components		Metrics			
	Recon	Contra	ACC	F1	NMI	ARI
InsectWingbeat	✓	✓	0.1344 ± 0.0023	0.1202 ± 0.0033	0.0153 ± 0.0012	0.0061 ± 0.0007
	✓	✓	0.1206 ± 0.0063	0.1104 ± 0.0111	0.0053 ± 0.0027	0.0025 ± 0.0014
	✓	✓	0.1000 ± 0.0000	0.1818 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
	✓	✓	0.1000 ± 0.0000	0.1818 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
JapaneseVowels	✓	✓	0.8647 ± 0.0447	0.8626 ± 0.0432	0.8195 ± 0.0260	0.7682 ± 0.0565
	✓	✓	0.8859 ± 0.0310	0.8942 ± 0.0270	0.8224 ± 0.0315	0.7627 ± 0.0522
	✓	✓	0.6441 ± 0.1450	0.6632 ± 0.1467	0.5833 ± 0.1473	0.4328 ± 0.1564
	✓	✓	0.1844 ± 0.0000	0.3113 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
LSST	✓	✓	0.2042 ± 0.0087	0.1805 ± 0.0063	0.1456 ± 0.0063	0.0555 ± 0.0044
	✓	✓	0.3362 ± 0.0017	0.2228 ± 0.0094	0.1699 ± 0.0142	0.0714 ± 0.0062
	✓	✓	0.1969 ± 0.0139	0.1776 ± 0.0102	0.1427 ± 0.0084	0.0537 ± 0.0075
	✓	✓	0.3155 ± 0.0000	0.4797 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
Libras	✓	✓	0.3156 ± 0.0374	0.3162 ± 0.0405	0.4101 ± 0.0302	0.1502 ± 0.0253
	✓	✓	0.3600 ± 0.0283	0.3572 ± 0.0231	0.4518 ± 0.0285	0.1916 ± 0.0259
	✓	✓	0.4161 ± 0.0210	0.4303 ± 0.0315	0.5402 ± 0.0261	0.2622 ± 0.0231
	✓	✓	0.2644 ± 0.0235	0.2673 ± 0.0222	0.3222 ± 0.0456	0.0927 ± 0.0307
MotorImagery	✓	✓	0.5323 ± 0.0147	0.5271 ± 0.0154	0.0039 ± 0.0034	0.0025 ± 0.0046
	✓	✓	0.5079 ± 0.0044	0.4726 ± 0.0036	0.0004 ± 0.0004	0.0016 ± 0.0004
	✓	✓	0.5270 ± 0.0074	0.5048 ± 0.0179	0.0034 ± 0.0023	0.0009 ± 0.0017
	✓	✓	OOM	OOM	OOM	OOM
NATOPS	✓	✓	0.4756 ± 0.0180	0.4638 ± 0.0129	0.4697 ± 0.0383	0.3103 ± 0.0385
	✓	✓	0.4972 ± 0.0478	0.4940 ± 0.0434	0.4796 ± 0.0594	0.3245 ± 0.0508
	✓	✓	0.4328 ± 0.0428	0.4242 ± 0.0431	0.3578 ± 0.0955	0.2377 ± 0.0650
	✓	✓	0.1967 ± 0.0600	0.3526 ± 0.1337	0.0770 ± 0.1539	0.0221 ± 0.0441
PEMS-SF	✓	✓	0.3905 ± 0.0724	0.4364 ± 0.0969	0.3625 ± 0.1185	0.2028 ± 0.0839
	✓	✓	0.2368 ± 0.0216	0.2476 ± 0.0355	0.1705 ± 0.0511	0.0609 ± 0.0218
	✓	✓	0.2700 ± 0.0273	0.2785 ± 0.0369	0.1930 ± 0.0781	0.0771 ± 0.0412
	✓	✓	0.1477 ± 0.0000	0.2574 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
PenDigits	✓	✓	0.6796 ± 0.0497	0.6724 ± 0.0488	0.6440 ± 0.0381	0.5249 ± 0.0488
	✓	✓	0.7139 ± 0.0446	0.7030 ± 0.0448	0.6472 ± 0.0277	0.5420 ± 0.0420
	✓	✓	0.1041 ± 0.0000	0.1885 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
	✓	✓	0.1041 ± 0.0000	0.1885 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
PhonemeSpectra	✓	✓	0.1125 ± 0.0040	0.1119 ± 0.0039	0.1765 ± 0.0084	0.0333 ± 0.0020
	✓	✓	0.0985 ± 0.0017	0.0967 ± 0.0012	0.1443 ± 0.0051	0.0243 ± 0.0012
	✓	✓	0.0256 ± 0.0000	0.0500 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
	✓	✓	0.0256 ± 0.0000	0.0500 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
RocketSports	✓	✓	0.5908 ± 0.0704	0.5918 ± 0.0700	0.3431 ± 0.0775	0.2754 ± 0.0886
	✓	✓	0.6389 ± 0.0424	0.6436 ± 0.0345	0.4214 ± 0.0229	0.3526 ± 0.0483
	✓	✓	0.4488 ± 0.0172	0.4449 ± 0.0185	0.1929 ± 0.0490	0.1187 ± 0.0352
	✓	✓	0.4244 ± 0.0367	0.3947 ± 0.0725	0.1550 ± 0.0234	0.1061 ± 0.0245
Self RegulationSCP1	✓	✓	0.7747 ± 0.0140	0.7738 ± 0.0136	0.2371 ± 0.0269	0.3014 ± 0.0296
	✓	✓	0.7458 ± 0.0132	0.7421 ± 0.0156	0.2011 ± 0.0274	0.2411 ± 0.0261
	✓	✓	0.7337 ± 0.0714	0.7246 ± 0.0859	0.1956 ± 0.0777	0.2376 ± 0.1041
	✓	✓	0.7480 ± 0.0175	0.7477 ± 0.0176	0.1878 ± 0.0261	0.2458 ± 0.0341
Self RegulationSCP2	✓	✓	0.5253 ± 0.0151	0.5162 ± 0.0079	0.0036 ± 0.0050	0.0010 ± 0.0042
	✓	✓	0.5347 ± 0.0139	0.5325 ± 0.0152	0.0041 ± 0.0021	0.0030 ± 0.0029
	✓	✓	0.5158 ± 0.0173	0.5106 ± 0.0157	0.0017 ± 0.0031	0.0003 ± 0.0040
	✓	✓	0.5200 ± 0.0061	0.5191 ± 0.0059	0.0013 ± 0.0007	0.0009 ± 0.0009
Spoken ArabicDigits	✓	✓	0.4260 ± 0.0202	0.4328 ± 0.0168	0.3868 ± 0.0165	0.2365 ± 0.0194
	✓	✓	0.3631 ± 0.0512	0.3642 ± 0.0468	0.3212 ± 0.0486	0.1523 ± 0.0453
	✓	✓	0.1000 ± 0.0000	0.1819 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
	✓	✓	0.1000 ± 0.0000	0.1819 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
StandWalkJump	✓	✓	0.4000 ± 0.0637	0.3740 ± 0.0724	0.0441 ± 0.0480	0.0171 ± 0.0522
	✓	✓	0.4593 ± 0.0378	0.4537 ± 0.0361	0.0530 ± 0.0194	0.0194 ± 0.0251
	✓	✓	0.4519 ± 0.0363	0.4355 ± 0.0497	0.0543 ± 0.0236	0.0178 ± 0.0199
	✓	✓	0.4296 ± 0.0181	0.3740 ± 0.0214	0.0924 ± 0.0506	0.0016 ± 0.0243
UWaveGesture Library	✓	✓	0.4164 ± 0.0435	0.4043 ± 0.0445	0.3456 ± 0.0479	0.1996 ± 0.0452
	✓	✓	0.4023 ± 0.0342	0.3992 ± 0.0359	0.2887 ± 0.0371	0.1684 ± 0.0280
	✓	✓	0.6318 ± 0.0456	0.6520 ± 0.0381	0.5779 ± 0.0160	0.4497 ± 0.0166
	✓	✓	0.1250 ± 0.0000	0.2222 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000

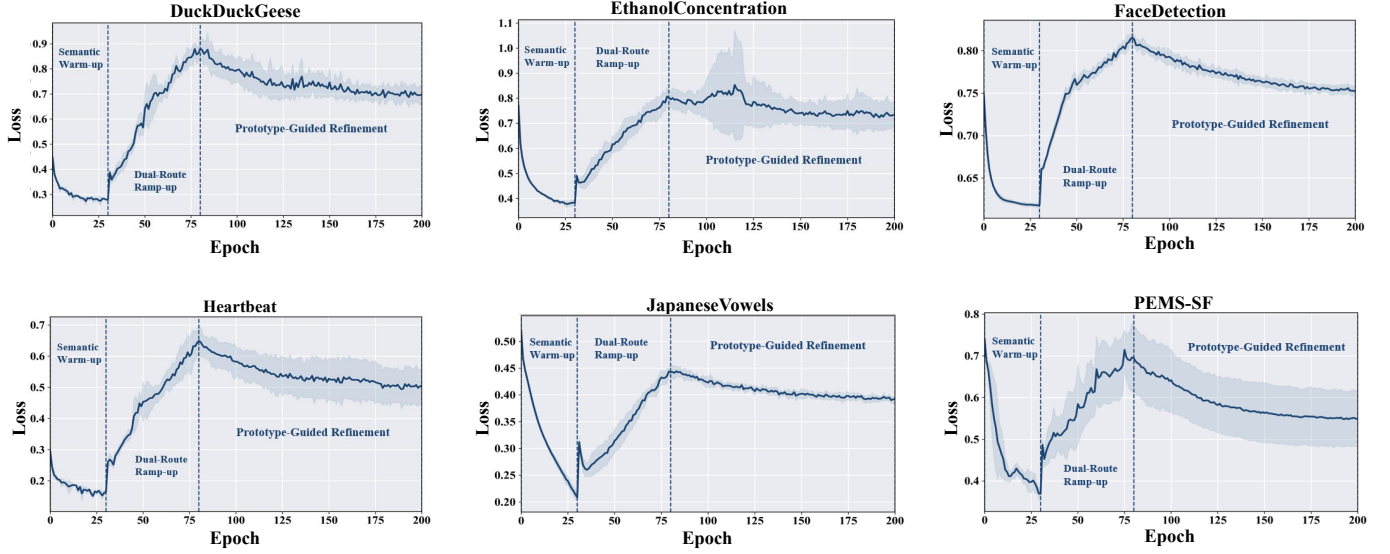


Fig. 1. Additional training-loss curves during SIGMA training on six UEA datasets. Each curve reports the mean loss across five random seeds, and the shaded band denotes standard deviation. Vertical dashed lines mark the semantic warm-up, dual-route ramp-up, and prototype-guided refinement stages.

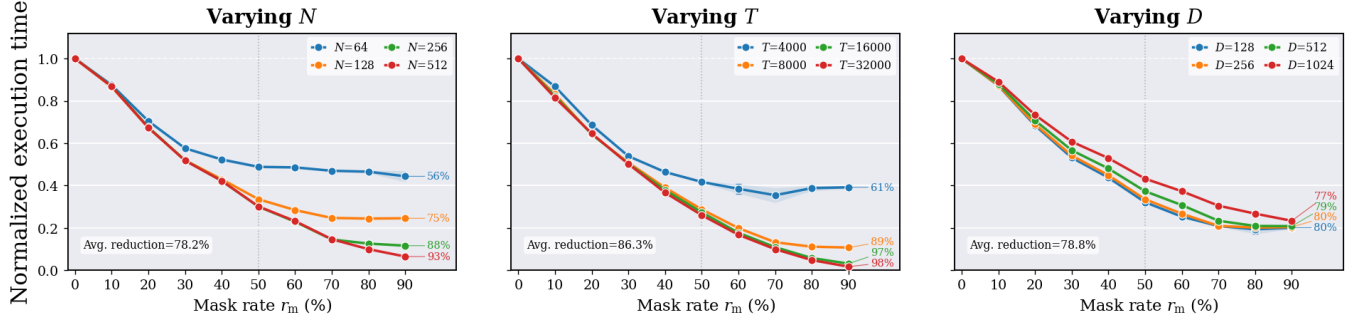


Fig. 2. Industrial-scale execution-time reduction under increasing mask rates. The three panels vary N , T , and D , respectively. For each setting, execution time is normalized by the cost at zero mask rate. Shaded bands denote standard deviation, and endpoint annotations report the reduction at 90% mask rate.

ment, instead of allowing early pseudo labels to dominate an under-organized latent space.

After the dual-route ramp-up begins, several curves show a temporary increase or fluctuation. The change is expected because the optimization target expands from reconstruction-centered semantic anchoring to inter-view complementarity and prototype-level alignment. Specifically, $\mathcal{L}_{\text{orth}}$ starts to reduce homogeneous view organization, and $\mathcal{L}_{\text{pa}}$ starts to pull fused embeddings toward cluster prototypes. A monotonic decrease at this boundary is therefore not required. The more important diagnostic is whether the loss remains controllable after the new objectives are introduced.

The later epochs provide the key stability signal for Q3. DuckDuckGeese, FaceDetection, Heartbeat, JapaneseVowels, and PEMS-SF enter a smoother refinement stage after the transition peak. EthanolConcentration shows a short high-variance interval, which is consistent with its weak cluster correspondence, but the curve also becomes stable in later epochs. Moreover, the shaded bands remain bounded during

prototype-guided refinement. Persistent oscillation or collapse would suggest unreliable pseudo-label feedback, yet Fig. 1 does not show such a pattern.

The additional curves therefore strengthen the stability conclusion in the main paper. Reconstruction first anchors the retained temporal content after partial observation. Inter-view orthogonality then promotes complementary endogenous views, and prototype alignment refines the fused cluster space after the representation has obtained a stable semantic basis. In this staged organization, semantic preservation and cluster refinement are introduced at different training phases and assigned distinct learning roles. As a result, SIGMA connects admission, restoration, and prototype-guided clustering without forcing redundancy suppression and semantic preservation into the same objective, which supports the decoupled prune-and-preserve design from an optimization perspective.

D. Industrial-Scale Efficiency under Increasing Mask Rates

The main paper answers Q3 by evaluating whether source-level admission improves scalability when sample size, sequence length, and input dimension increase. The efficiency comparison in the main paper shows the external behavior across methods. This appendix subsection further asks whether the efficiency gain follows the intended mechanism inside SIGMA. If the main cost reduction comes from admission before encoder-side feature interaction, then excluding more patches should reduce normalized execution time in a predictable way. Fig. 2 examines this mechanism under varying N , T , and D , which correspond to sample-side workload, sequence length, and input dimension.

The mask rate is determined by the retained patch ratio. Given patch length ℓ_p , each input sequence is divided into $m = \lceil t/\ell_p \rceil$ patches. Under retained ratio r_a , Confidence-Calibrated Masked Admission keeps $m_a = \lceil r_a m \rceil$ patches for encoder-side feature interaction. The corresponding mask rate is calculated as:

$$r_m = 1 - \frac{m_a}{m}. \quad (1)$$

In this experiment, r_m increases from 0% to 90% by decreasing r_a . A zero mask rate means that all patches participate in the encoder. A larger mask rate means that more background-dominated patches are removed before attention computation. For each scale setting, execution time is normalized by the cost at zero mask rate, which makes the three scale factors comparable from a relative-cost perspective.

Fig. 2 shows a consistent decrease in normalized execution time as the mask rate increases. The average runtime reduction reaches 78.2% under varying N , 86.3% under varying T , and 78.8% under varying D . The reduction is most pronounced when sequence length increases. This result agrees with the complexity analysis in the main paper, where the dominant attention term depends on the retained patch number m_a after admission. A larger r_m reduces m_a , and a smaller retained patch set produces fewer pairwise interactions inside the encoder. Therefore, the measured runtime gain comes from input-side exclusion before redundant patches can shape feature interaction.

The three scale groups also clarify how the same admission mechanism behaves under different workloads. When N increases, the same patch admission rule is applied to more samples, so the accumulated saving becomes clearer. When T increases, the benefit grows more strongly because longer sequences create more candidate patches and enlarge the full-patch attention space. When D increases, the runtime reduction remains stable because fewer admitted tokens go through high-dimensional projection and attention computation. These observations show that CCMA provides computation-aware pruning at the source. That is, CCMA does not merely assign smaller weights to redundant content after full encoding. It changes the admissible input set and directly reduces the dominant representation-learning cost.

Overall, the industrial-scale efficiency results complement the main-paper answer to Q3. The main comparison shows

that SIGMA scales favorably against competing methods, and Fig. 2 explains why the scaling advantage appears. Increasing the mask rate reduces execution time by suppressing redundant patch participation before encoder-side interaction, whereas routed objectives keep the retained content usable for clustering after partial observation. The result supports the prune-and-preserve principle of SIGMA, where source-level redundancy suppression, semantic preservation, and prototype-guided cluster refinement are assigned distinct roles within one compact clustering framework.

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